14.8 Solution: From Eq. (12.1), the equation fo motion for a particle with mass m and charge ze in an electric field is

$$\frac{d}{dt}\left(\gamma mc\boldsymbol{\beta}\right) = \gamma^3 mc\boldsymbol{\beta}\left(\boldsymbol{\beta}\cdot\dot{\boldsymbol{\beta}}\right) + \gamma mc\dot{\boldsymbol{\beta}} = ze\mathbf{E}.$$

Dot by $\dot{\boldsymbol{\beta}}$ on both sides, we will have

$$\gamma^3 mc\beta^2\left(\boldsymbol{\beta}\cdot\dot{\boldsymbol{\beta}}\right) + \gamma mc\left(\boldsymbol{\beta}\cdot\dot{\boldsymbol{\beta}}\right) = \gamma mc(\gamma^2\beta^2+1)\left(\boldsymbol{\beta}\cdot\dot{\boldsymbol{\beta}}\right) = \gamma^3 mc\left(\boldsymbol{\beta}\cdot\dot{\boldsymbol{\beta}}\right) = ze\boldsymbol{\beta}\cdot\mathbf{E}.$$

Therefore,

$$\boldsymbol{\beta}\cdot\dot{\boldsymbol{\beta}} = \frac{ze}{\gamma^3 mc}\boldsymbol{\beta}\cdot\mathbf{E},$$

and

$$\dot{\boldsymbol{\beta}} = \frac{1}{\gamma mc}\left(ze\mathbf{E} - \gamma^3 mc\boldsymbol{\beta}\left(\boldsymbol{\beta}\cdot\dot{\boldsymbol{\beta}}\right)\right) = \frac{ze}{\gamma mc}\left(\mathbf{E} - \boldsymbol{\beta}\left(\boldsymbol{\beta}\cdot\mathbf{E}\right)\right).$$

Using Lamor's formula, Eq. (14.26), the radiated power is

$$P(t) = \frac{2}{3}\frac{z^2e^2}{c}\gamma^6\left[\left|\dot{\boldsymbol{\beta}}\right|^2 - \left|\boldsymbol{\beta}\times\dot{\boldsymbol{\beta}}\right|^2\right].$$

Since

$$\left|\dot{\boldsymbol{\beta}}\right|^2 = \frac{z^2e^2}{\gamma^2m^2c^2}\left[\left|\mathbf{E}\right|^2 - 2\left|\boldsymbol{\beta}\cdot\mathbf{E}\right|^2 + \beta^2\left|\boldsymbol{\beta}\cdot\mathbf{E}\right|^2\right],$$

and

$$\left|\boldsymbol{\beta}\times\dot{\boldsymbol{\beta}}\right|^2 = \frac{z^2e^2}{\gamma^2m^2c^2}\left|\boldsymbol{\beta}\times\mathbf{E}\right|^2 = \frac{z^2e^2}{\gamma^2m^2c^2}\left[\beta^2\left|\mathbf{E}\right|^2 - \left|\boldsymbol{\beta}\cdot\mathbf{E}\right|^2\right],$$

then

$$\begin{aligned} P(t) &= \frac{2}{3}\frac{z^4e^4}{m^2c^3}\gamma^4\left[\left|\mathbf{E}\right|^2 - 2\left|\boldsymbol{\beta}\cdot\mathbf{E}\right|^2 + \beta^2\left|\boldsymbol{\beta}\cdot\mathbf{E}\right|^2 - \beta^2\left|\mathbf{E}\right|^2 + \left|\boldsymbol{\beta}\cdot\mathbf{E}\right|^2\right] \\ &= \frac{2}{3}\frac{z^4e^4}{m^2c^3}\gamma^4(1-\beta^2)\left[\left|\mathbf{E}\right|^2 - \left|\boldsymbol{\beta}\cdot\mathbf{E}\right|^2\right] \\ &= \frac{2}{3}\frac{z^4e^4}{m^2c^3}\gamma^2\left[\left|\mathbf{E}\right|^2 - \left|\boldsymbol{\beta}\cdot\mathbf{E}\right|^2\right]. \end{aligned}$$

This can also be obtained from Prob. 14.11 (a), with $\mathbf{B} = 0$.

Assuming straight path of the particle, we have

$$\left|\mathbf{E}\right| = \frac{Ze}{x^2+b^2}, \quad \mathbf{E}\cdot\boldsymbol{\beta} = -\frac{Ze}{x^2+b^2}\frac{\beta x}{\sqrt{x^2+b^2}},$$

and

$$P(t) = \frac{2}{3}\frac{z^4Z^2e^6}{m^2c^3}\gamma^2\frac{1}{(x^2+b^2)^2}\left(1 - \frac{\beta^2x^2}{x^2+b^2}\right).$$

Integrating the power, and notice that

$$dt = \frac{dx}{v} = \frac{dx}{c\beta},$$

the total energy radiate is then

$$\Delta W = \int_{-\infty}^{+\infty}P(t)dt = \frac{2}{3}\frac{z^4Z^2e^6}{m^2c^4\beta}\gamma^2\int_{-\infty}^{+\infty}\left[\frac{1}{(x^2+b^2)^2} - \frac{\beta^2x^2}{(x^2+b^2)^3}\right]dx$$

$$
\begin{aligned}
&= \frac{2}{3}\frac{z^4Z^2e^6}{m^2c^4\beta}\gamma^2\left[\frac{\pi}{2b^3}-\frac{\pi\beta^2}{8b^3}\right]=\frac{2}{3}\frac{\pi z^4Z^2e^6}{m^2c^4\beta}\gamma^2\left[\frac{3}{8}+\frac{1-\beta^2}{8}\right]\frac{1}{b^3}\\
&= \frac{2}{3}\frac{\pi z^4Z^2e^6}{m^2c^4\beta}\gamma^2\left[\frac{3}{8}+\frac{1}{8\gamma^2}\right]\frac{1}{b^3}=\frac{\pi z^4Z^2e^6}{4m^2c^4\beta}\left(\gamma^2+\frac{1}{3}\right)\frac{1}{b^3}.
\end{aligned}
$$

For non-relativistic motion, $v_0 = c\beta$ and $\gamma \to 1$, this reduces to the result of Prob. 14.7.